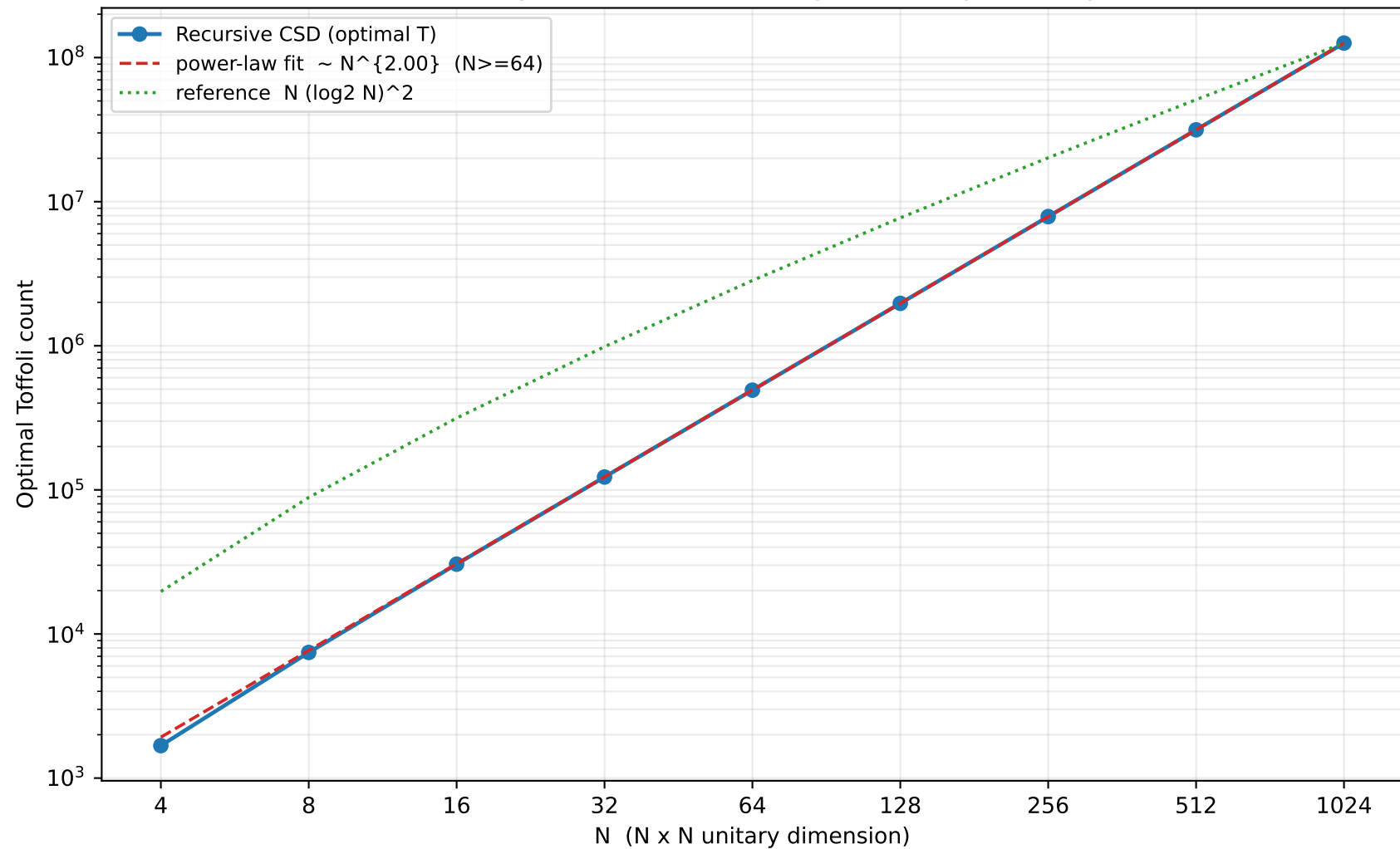


Recursive CSD synthesis: Toffoli scaling vs N (single unitary, b=32)



Recursive CSD optimal Toffoli vs N (b=32); fit $T \sim N^{2.00}$

N	Toffoli	T / N	T / (N log ₂ ² N)
4	1,680	420.0	105.00
8	7,446	930.8	103.42
16	30,522	1,907.6	119.23
32	122,850	3,839.1	153.56
64	492,210	7,690.8	213.63
128	1,969,746	15,388.6	314.05
256	7,880,082	30,781.6	480.96
512	31,521,810	61,566.0	760.07
1024	126,089,106	123,133.9	1231.34